

Fundamental of AI

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Unit – 2

Knowledge Representation - Natural Language Processing

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Outline

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- Importance of NLP
- Components of NLP
- Basic steps in NLP
- Challenges in NLP
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- Summary

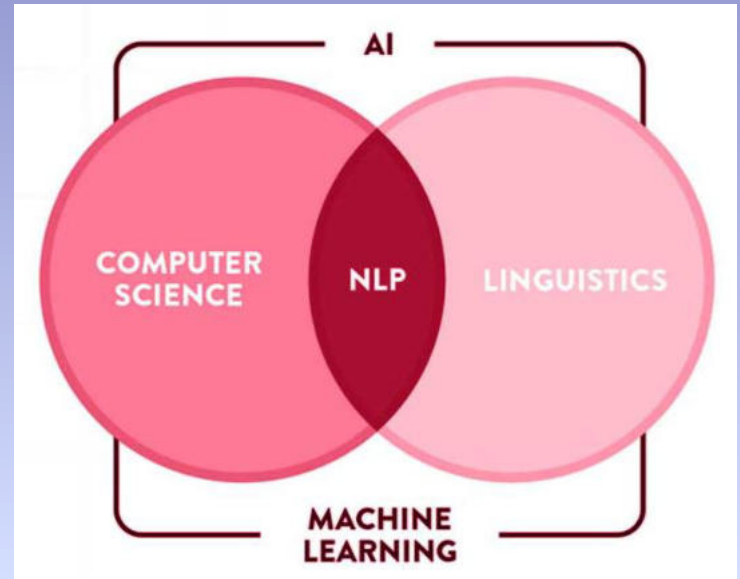
What is NLP?

Definition:

- Natural Language Processing (NLP) is a subfield of artificial intelligence (AI) that focuses on the interaction between computers and human language.
- It enables machines to understand, interpret, and respond to human language in a way that is both meaningful and useful.

Key Points:

- Combines linguistics (language) and computer science (algorithms).
- NLP allows computers to process and analyze vast amounts of natural language data.



Importance of NLP

- NLP uses many different techniques to enable computers to understand natural language as humans do.
- Whether the language is spoken or written, NLP can use AI to take real-world input, process it and make sense of it in a way a computer can understand.



- Just as humans have different sensors -- such as ears to hear and eyes to see -- computers have programs to read and microphones to collect audio.
- And just as humans have a brain to process that input, computers have a program to process their respective inputs.
- At some point in processing, the input is converted to code that the computer can understand.

Components of NLP

1. Syntax:

- Deals with the structure of sentences (grammar rules).
- Involves parsing to understand how words relate in a sentence.

2. Semantics:

- Focuses on the meaning of words and sentences.
- Aims to understand the context behind words and their relationships.

3. Pragmatics:

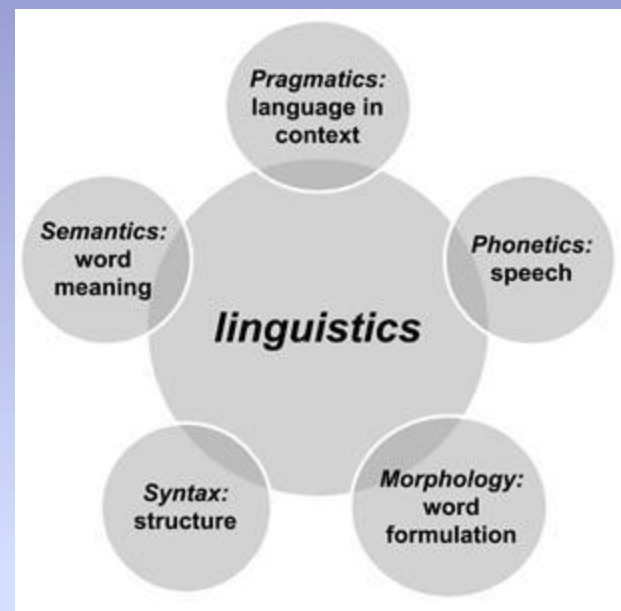
- Looks at how language is used in context and how meaning changes in different situations.

4. Morphology:

- Studies the structure and formation of words.

5. Phonetics:

- Involves the study of sounds in speech, including speech-to-text conversion.



Basic Steps in NLP

1. Tokenization:

- Breaking text into smaller units (tokens), such as words, sentences, or paragraphs.



Basic Steps in NLP

2. Named Entity Recognition (NER):

- Identifying proper nouns or named entities (e.g., names of people, organizations, locations).

ORGANISATION LOCATION DATE PERSON WEAPON

The **ISIS** ORG has claimed responsibility for a suicide bomb blast in the **Tunisian** LOC capital **earlier this week** DATE, the **militant group** ORG's **Amaq news agency** ORG said on **Thursday** DATE. A **militant** PER wearing an **explosives belt** WEAPON blew himself up in **Tunis** LOC

Basic Steps in NLP (Contd..)

3. Part-of-Speech Tagging:

- Assigning words with a part of speech (noun, verb, adjective, etc.).

4. Sentiment Analysis:

- Determining whether the text expresses positive, negative, or neutral sentiment.

SENTIMENT ANALYSIS



POSITIVE

"Great service for an affordable price.
We will definitely be booking again."



NEUTRAL

"Just booked two nights
at this hotel."



NEGATIVE

"Horrible services. The room
was dirty and unpleasant.
Not worth the money."

Basic Steps in NLP (Contd..)

5. Stemming and Lemmatization:

- Reducing words to their base or root form (e.g., “running” to “run”).

Stemming vs Lemmatization

Stemming

achieve -> achiev
achieving -> achiev

- Can reduce words to a stem that is not an existing word
- Operates on a single word without knowledge of the context
- Simpler and faster

Lemmatization

achieve -> achieve
achieving -> achieve

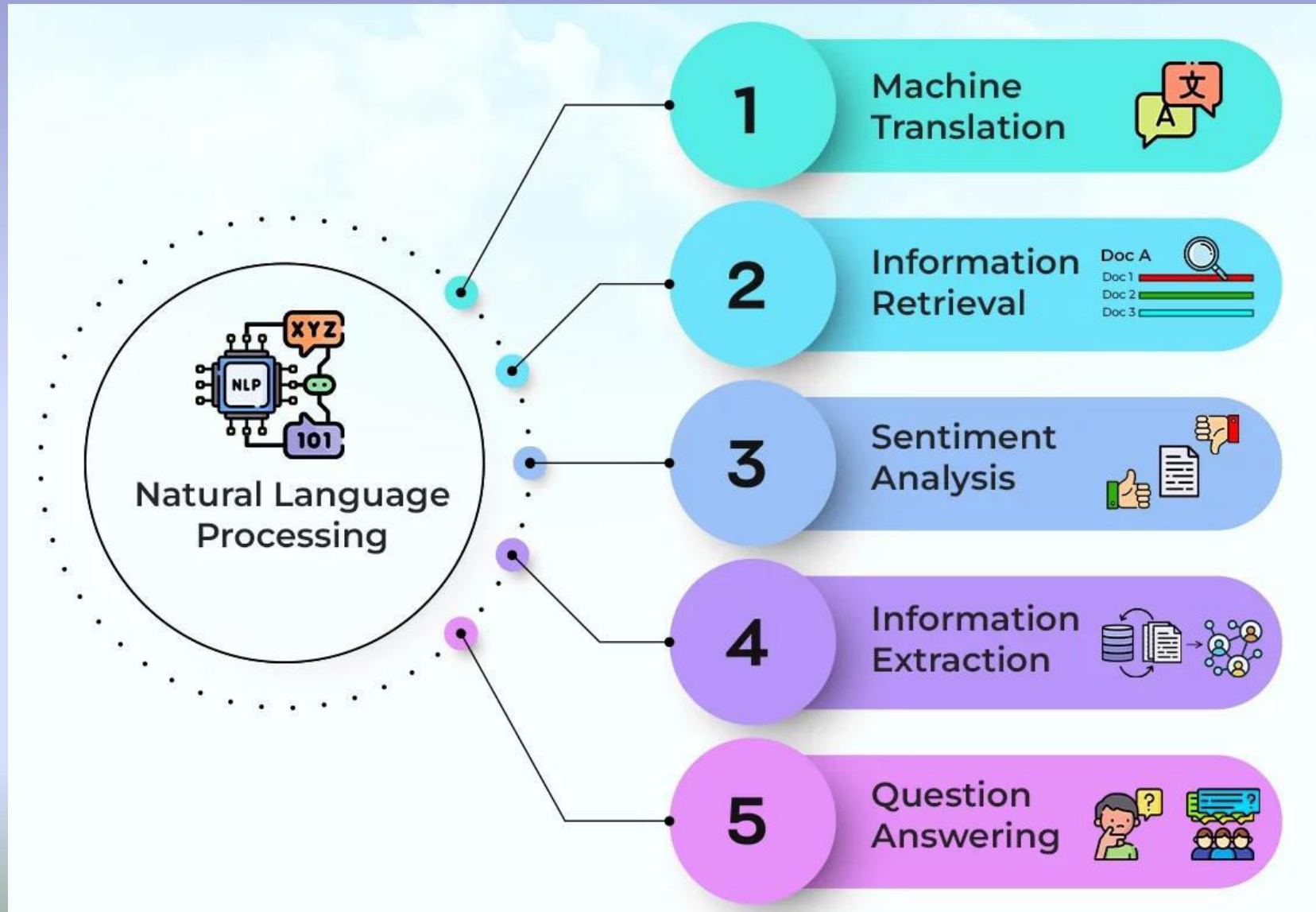
- Reduces inflected words to their lemma, which is always an existing word
- Can leverage context to find the correct lemma of a word
- More accurate but slower

Challenges in NLP

- **Ambiguity:** Words can have multiple meanings depending on context (e.g., "bat" can mean a flying mammal or a piece of sports equipment).
- **Syntax Variations:** Sentence structures vary across languages and dialects, making it difficult for machines to interpret.
- **Context Understanding:** Understanding the context of a conversation is crucial for accurate interpretation, especially in longer texts or dialogues.
- **Sarcasm and Irony:** Machines struggle to understand nuances like sarcasm, making interpretation difficult.

<u>SARCASM</u>	<u>VS</u>	<u>IRONY</u>
• It is used intentionally or willingly • You plan it after some amount of thinking		• It is usually an unconscious effort • You don't plan the circumstances that lead to the ironic situation

Applications of NLP



Applications of NLP

Applications of NLP in Communication



Applications of NLP

- **Machine Translation:** Automatically translating text or speech between languages.
- **Speech Recognition:** Converting spoken language into text for virtual assistants, transcription services, etc.
- **Text Classification:** Categorizing text into predefined categories (e.g., spam detection, news categorization).
- **Chatbots and Virtual Assistants:** Enabling interactive conversations with users (e.g., Siri, Alexa, Google Assistant).
- **Sentiment Analysis:** Analyzing customer reviews or social media posts to determine sentiment (positive/negative).
- **Information Retrieval:** Identifying and retrieving the data from the database
- **Information Extraction:** Automatically extract structured information from unstructured text

Current Trends of NLP

Deep Learning in NLP:

- Using neural networks (e.g., transformers) for more accurate language modeling and text generation.
- Models like GPT-3 and BERT are pushing the boundaries of NLP capabilities.

Multilingual NLP:

- Building models that can handle multiple languages and dialects seamlessly.

Explainability in NLP Models:

- Understanding why a model made a specific decision or output, which is important for trust in AI systems.

Future of NLP

Enhanced Human-Computer Interaction:

- More natural, seamless communication between humans and machines.

Improved Contextual Understanding:

- Achieving deeper understanding of complex sentences, idioms, and cultural nuances.

Personalized Services:

- Using NLP to create more tailored and intuitive user experiences, such as personalized news feeds or virtual assistants.

Summary

- NLP enables machines to interact with human language, making it one of the most exciting areas of AI research.
- While challenges like ambiguity and context understanding remain, techniques such as deep learning and advancements in multilingual processing are driving the future of NLP.